

University of Botswana
Department of Computer Science
Tutorial - Sets

Question 1

Show that if A , B , and C are sets, then $(A \cap B \cap C)' = A' \cup B' \cup C'$ by showing that each side is a subset of the other side.

Question 2

Let A be a set. Show that $A \cup \emptyset = A$ by showing that:

- $A \cup \emptyset \subseteq A$
- $A \subseteq A \cup \emptyset$

Question 3

Prove that if A and B are sets, then $(A \cap A) = A$ by showing that:

- $(A \cap A) \subseteq A$

Question 4

Prove that if A and B are sets, then $(\emptyset - A) = \emptyset$ by showing that:

- $(\emptyset - A) \subseteq \emptyset$

Question 5

Prove that if A and B are sets, then $A \cup (A \cap B) = A$ by showing that:

- $A \cup (A \cap B) \subseteq A$

Question 6

Prove that if A and B are sets, then $(A \cup (B - A)) = (A \cup B)$ by showing that:

- $(A \cup (B - A)) \subseteq (A \cup B)$
- $(A \cup B) \subseteq (A \cup (B - A))$